

Nathan Perez

956-378-3100 | nperez1018@gmail.com | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Texas Rio Grande Valley
Bachelor of Science in Computer Science

Edinburg, TX
2021 – 2025

TECHNICAL SKILLS

Languages: SQL, Linux, C++, Python, Java, R

Tools: MySQL, Excel, Tableau, Microsoft 365, Pandas, Git, Wireshark, Azure,

Soft Skills: Problem-Solving, Quick Learner, Communication, Self-Sufficient, Teamwork, Organization

PROJECTS

Academic Performance Trend Analysis (SQL, MySQL, Tableau, Excel)

- Analyzed structured datasets simulating academic environments, including class performance, course drop rates, and reasons for student attrition
- Queried data using SQL to identify trends, patterns, and factors influencing outcomes
- Created dashboards and reports to visualize and communicate findings
- Performed data cleaning and transformation to prepare datasets for analysis

Phishing URL Detection Analysis (Python, Scikit-learn, Random Forest, XGBoost, pandas, Google Colab)

- Developed and evaluated Random Forest and XGBoost models to detect phishing URLs using a dataset of 235k+ instances
- Performed a comparative analysis using accuracy, precision, recall, and F1-score to assess model performance
- Conducted data preprocessing to clean and organize the data to utilize feature analysis to improve model reliability and model interpretability

Voyage: Travel Itinerary Generator (Figma, HTML/CSS, Python, Flask, PostgreSQL, APIs, GitHub, JavaScript)

- Developed a full-stack web application that generates personalized travel itineraries based on user inputs (budget, location, dates, preferences)
- Integrated multiple APIs including Google Maps, Open-Meteo, Unsplash, and Gemini to enhance data-driven itinerary generation
- Designed and managed backend data handling using the Flask framework and PostgreSQL for database management
- Collaborated using GitHub workflows and designed UI/UX prototypes in Figma

OUTSIDE EXPERIENCE

DoD Cyber Sentinel Challenge (2025)

- Participated in a Department of Defense cybersecurity competition designed to simulate real-world challenges in fields such as forensics, OSINT, networking, malware analysis, and web security.
- Ranked approximately ~450 out of ~2000 participants
- Engaged with real-world cybersecurity challenges and developed an understanding of OSINT, networking, and system vulnerabilities.